

University of Nottingham Malaysia

SCHOOL OF COMPUTER SCIENCE
A LEVEL 1 MODULE, AUTUMN SEMESTER 2022-2023

COMPUTER FUNDAMENTALS (COMP1027)

Time allowed: **ONE** Hour

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer ALL questions

Marks available for sections of questions are shown in brackets in the right-hand margin

No calculators are permitted in this examination

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used

DO NOT turn your examination paper over until instructed to do so

ADDITIONAL MATERIAL:

None

INFORMATION FOR INVIGILATORS:

Please collect the exam paper at the end of the exam

1. Number Representation, Binary Arithmetic & Digital Logic [30%]

Table 1.

Decimal	Binary	1's Complement	2's Complement
1111	<i>a</i>	<i>b</i>	<i>c</i>
<i>f</i>	<i>e</i>	<i>d</i>	1111 1101 1011 1010

- a) Given Table 1, calculate the 6 values labelled *a-f*. Your representations for Binary, 1's Complement and 2's Complement must be in 16-bits width.

[6 marks]

- b) Given the following values of *x*, *y*, and *z* below (in Decimal numbers):

- $x = -2051$,
- $y = 1029$,
- $z = -517$

Calculate, showing your workings in detail, the binary representation in 16-bit TWO's complement.

[6 marks]

- c) Using your **16-bits** TWO's complement answers in 1(b), calculate the following sums. Show your workings in detail. Your answer should be in 16-bits width.

i. $x + y$

[2 marks]

ii. $y - z$

[2 marks]

iii. $x + z$

[2 marks]

- d) AND, OR and NOT are known as **basic** gates. However, they can be represented by **universal** gate(s). Please draw AND, OR and NOT logic diagrams using relevant **universal** gate(s) with A, B as inputs and Y as output. Please clearly label the inputs and output in your logic diagram.

[12 marks]

2. Boolean Algebra & ALU [40%]

- a) Tessa is a talented baker who wishes to open her own café in 2023. In order to realise her dreams, she needs to secure a location whereby her café can be open, purchase relevant furniture, enlist several helpers, and prepare a set of delightful choices for the customers to choose from. For the short-term, she will be happy to successfully fulfil two out of the four conditions, namely a location and menu.

- i. Generate a **truth table** to represent the conditions that Tessa needs to fulfil to ensure her café is successfully operational by February 2023. Please represent location, furniture, helpers and menu as A, B, C and D in your truth table.

[2 marks]

- ii. Based on the truth table in 2a(i), write out the expression (pre-simplified).

[1 mark]

P.T.O

Identity Name	AND Form	OR Form
Identity Law	$1x = x$	$0+x = x$
Null (or Dominance) Law	$0x = 0$	$1+x = 1$
Idempotent Law	$xx = x$	$x+x = x$
Inverse Law	$x\bar{x} = 0$	$x+\bar{x} = 1$
Commutative Law	$xy = yx$	$x+y = y+x$
Associative Law	$(xy)z = x(yz)$	$(x+y)+z = x+(y+z)$
Distributive Law	$x+yz = (x+y)(x+z)$	$x(y+z) = xy+xz$
Absorption Law	$x(x+y) = x$	$x+xy = x$
DeMorgan's Law	$\overline{(xy)} = \bar{x}+\bar{y}$	$\overline{(x+y)} = \bar{x}\bar{y}$
Double Complement Law	$\overline{\bar{x}} = x$	

Extracted from: L. Null, L. and J. Lobur, The essentials of computer organisation and architecture, Jones & Bartlett Publishers, 2003 (Fig. 1.3, P.26)

Figure 1.

- iii. Using the given Basic Identities of Boolean Algebra in Figure 1, simplify your expression in 2a(ii). Detail each step. **[13 marks]**
- iv. Based on your answer in 2a(iii), draw the Logic Circuit Diagram to implement the circuit (using the relevant combinations of ONLY basic gates). **[1 mark]**
- b) Figure 2 illustrates an ALU design with **vital components missing**. Please note that inputs a , b , c , and d are 16-bit values.

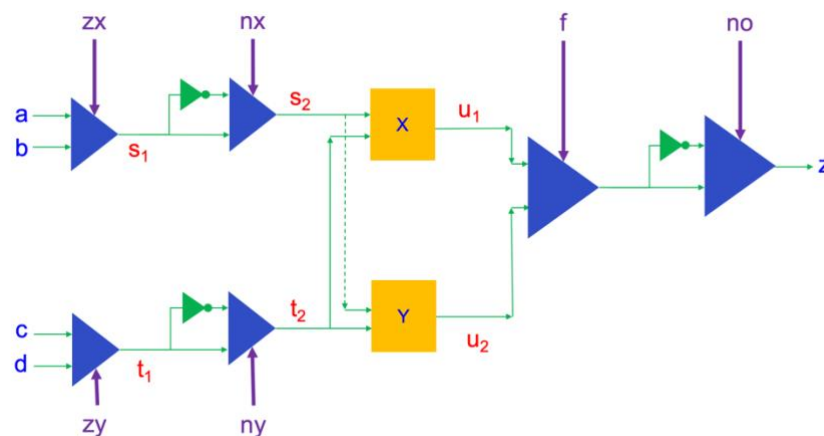


Figure 2.

- (i). Please identify and list the relevant component(s) labelled (X) & (Y). Your component(s) must be 16-bit wide gates. **[2 marks]**
- (ii). Once labels X & Y have been successfully identified, show (with full working details) how Figure 2 can achieve each of the listed output functionalities (z) according to the given corresponding control values in Table 2, below:

Table 2.

	zx	nx	zy	ny	f	no	z
m.	1	1	0	0	0	1	$a - 1$
n.	1	1	0	0	1	1	$-c$
o.	0	0	0	1	1	1	$c - a$

Hint: $-A = A\sim + 1$ { where $A\sim$ is the bitwise Not(A) }.

In your working details, you need to state what each of the variables (s_1 , s_2 , t_1 , t_2 , u_1 , u_2 , z) represents/contains in each case (m, n, o, in the table above).

[21 marks]

P.T.O

3. CPU & Networks [30%]

Table 3.

Address Range (Binary)	Purpose	Select Signal
$0000_2 - 0011_2$	ROM	ROMS
$0100_2 - 0111_2$	I/O	IOS
$1000_2 - 1011_2$	RAM1	RAM1S
$1100_2 - 1111_2$	RAM2	RAM2S

- a) A simple educational 4-bit computer is designed to have a small memory containing 16 locations, addressed by 4-bit address bus ($A_3 A_2 A_1 A_0$). The memory space is split and allocated as per Table 3. The I/O space has allocated ONE address for the Keyboard (KBD), which is address 0111_2 . The rest of the I/O space is currently unused.

Using a combination of the **basic gates only**, you are required to develop the following “select signals” in accordance with Table 3:

- The select signal ROMS, to be TRUE only when the address bus contains an address within the specified range for the ROM. **[3 marks]**
 - The select signal IOS, to be TRUE only when the address bus contains an address within the specified range for the I/O. **[3 marks]**
 - The select signal RAM1S, to be TRUE only when the address bus contains an address within the specified range for the RAM1. **[3 marks]**
 - The select signal RAM2S, to be TRUE only when the address bus contains an address within the specified range for the RAM2. **[3 marks]**
 - The select signal KBDS, to be TRUE only when the address bus contains the *specific* address of the keyboard location (KBD), which is address 0111_2 . **[3 marks]**
 - The select signal INVS, to be TRUE only when the address bus contains an address within the specified range of the I/O space, BUT the address is not the keyboard address. This would indicate an INVALID address, as the rest of the I/O space is currently unused. **[3 marks]**
- b) Computers that have 16-bytes word length or more, i.e., multiple-bytes words, such as the Hack Computer are required to use a unified approach in dealing with multiple-bytes words when communicating with other computers over a network. Justify why and detail the two possible approaches namely: *big-endian* and *little-endian*. **[10 marks]**
- c) Give examples of two protocols at layer three of the OSI network. **[2 marks]**

END